

Math 344, F19
Exam 2

Name: *Answer*

Instructions:

- There are totally 5 problems.
- You must show all of your work to receive a credit for a correct answer.
- No calculator is allowed in the quiz and exam.

Problem	Points	Score
1	24	
2	20	
3	20	
4	18	
5	18	
Total	100	

Good Luck !

Problem 1. (24 points) Let $A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ 0 & 0 & -2 & -4 \end{bmatrix}$.

(a) Transform A into the reduced echelon form R .

$$A = \begin{bmatrix} 1 & 3 & 1 & 2 \\ 2 & 6 & 4 & 8 \\ 0 & 0 & -2 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & -2 & -4 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\rightarrow \begin{bmatrix} 1 & 3 & 1 & 2 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \rightarrow \begin{bmatrix} 1 & 3 & 0 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} = R$$

pivot variables: x_1, x_3
free variables: x_2, x_4

(b) Find the dimension and a basis for the column space of A , $C(A)$.

$$\dim(C(A)) = 2$$

$$\text{Basis: } \left\{ \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 4 \\ -2 \end{bmatrix} \right\}$$

(c) Find the dimension and a basis for the row space of A , $C(A^T)$.

$$\dim(C(A^T)) = 2$$

$$\text{Basis: } \left\{ \begin{bmatrix} 1 \\ 3 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 6 \\ 4 \\ 8 \end{bmatrix} \right\} \quad \text{or} \quad \left\{ \begin{bmatrix} 1 \\ 3 \\ 0 \\ 2 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 2 \end{bmatrix} \right\}$$

(d) Find the dimension and a basis for the nullspace of A , $N(A)$.

$$\dim(N(A)) = 4 - 2 = 2$$

$$RX = 0 \Rightarrow \begin{cases} x_1 + 3x_2 = 0 \\ x_3 + 2x_4 = 0 \end{cases}$$

$$\text{Set } x_2 = 1, x_4 = 0, \text{ we get } S_1 = (-3, 1, 0, 0)$$

$$x_2 = 0, x_4 = 1, \text{ we get } S_2 = (0, 0, -2, 1)$$

$$\Rightarrow \text{Basis: } \left\{ \begin{bmatrix} -3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ -2 \\ 1 \end{bmatrix} \right\}$$

(e) Find the dimension and a basis for the left nullspace of A , $N(A^T)$.

$$\dim(N(A^T)) = 3 - 2 = 1$$

$$A^T y = 0 \Rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 3 & 6 & 0 \\ 1 & 4 & -2 \\ 2 & 8 & -4 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = 0 \Rightarrow \begin{bmatrix} 1 & 2 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = 0$$

$$\Rightarrow \begin{cases} y_1 + 2y_2 = 0 \\ 2y_2 - 2y_3 = 0 \end{cases} \Rightarrow y_1 = -2y_2 \Rightarrow y = y_2 \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} = c \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}$$

$$\Rightarrow \text{Basis: } \left\{ \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix} \right\}$$

Problem 2. (20 points)

(a) Find the dimension and a basis for the subspace of all 3×3 symmetric matrices.

$$\text{dimension} = 6$$

$$\text{Basis: } \left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{bmatrix}, \right. \\ \left. \begin{bmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} \right\}$$

(b) Construct a matrix $\mathbf{A}$ such that its column space $C(\mathbf{A})$ contains $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$, and its row space $C(\mathbf{A}^T)$ contains $\begin{bmatrix} 1 \\ 2 \end{bmatrix}$ and $\begin{bmatrix} 2 \\ 5 \end{bmatrix}$.

$$\mathbf{A} = \begin{bmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \end{bmatrix}$$

(c) Does it always hold $\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = \begin{vmatrix} e-d & e & f \\ b-a & b & c \\ h-g & h & i \end{vmatrix}$? Please justify your answer.

Yes.

$$\begin{vmatrix} a & b & c \\ d & e & f \\ g & h & i \end{vmatrix} = -\begin{vmatrix} d & e & f \\ a & b & c \\ g & h & i \end{vmatrix} = \begin{vmatrix} -d & e & f \\ -a & b & c \\ -g & h & i \end{vmatrix} = \begin{vmatrix} e-d & e & f \\ b-a & b & c \\ h-g & h & i \end{vmatrix}$$

Problem 3. (20 points)

(a) Find the straight line $y = C + Dt$ that comes closest to the values $\mathbf{b} = 0, 8, 8, 20$ at times $t = 0, 1, 3, 4$ by the least squares fitting.

$$A = \begin{bmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 3 \\ 1 & 4 \end{bmatrix}, \quad x = \begin{bmatrix} C \\ D \end{bmatrix}, \quad b = \begin{bmatrix} 0 \\ 8 \\ 8 \\ 20 \end{bmatrix} \rightarrow Ax = b$$

$$A^T A = \begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix}, \quad A^T b = \begin{bmatrix} 36 \\ 112 \end{bmatrix}$$

$$\text{Solve } A^T A \hat{x} = A^T b$$

$$\begin{bmatrix} 4 & 8 \\ 8 & 26 \end{bmatrix} \begin{bmatrix} C \\ D \end{bmatrix} = \begin{bmatrix} 36 \\ 112 \end{bmatrix} \Rightarrow \begin{cases} C = 1 \\ D = 4 \end{cases}$$

$$\Rightarrow \text{The least squares fitting line is} \\ y = 1 + 4t$$

(b) Let $A = \begin{bmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{bmatrix}$. Find A^{-1} using the cofactor formula $C^T / \det A$.

$$\det A = \begin{vmatrix} 0 & 1 & 3 \\ 1 & 0 & 1 \\ 2 & 1 & 1 \end{vmatrix} = (-1) \cdot \begin{vmatrix} 1 & 1 \\ 2 & 1 \end{vmatrix} + 3 \begin{vmatrix} 1 & 0 \\ 2 & 1 \end{vmatrix} \\ = (-1) \cdot (-1) + 3 \cdot 1 = 4.$$

$$C = \begin{bmatrix} -1 & 1 & 1 \\ 2 & -6 & 2 \\ 1 & 3 & -1 \end{bmatrix}$$

$$\Rightarrow A^{-1} = \frac{C^T}{\det A} = \frac{1}{4} \begin{bmatrix} -1 & 2 & 1 \\ 1 & -6 & 3 \\ 1 & 2 & -1 \end{bmatrix}$$

Problem 4. (18 points)

Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 1 & -1 & 0 \\ 2 & 0 & 4 \end{bmatrix}$. Find an orthonormal basis q_1, q_2, q_3 for the column space of A using the Gram-Schmidt process.

$$\tilde{q}_1 = a_1 = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \quad \|\tilde{q}_1\|^2 = 1^2 + 1^2 + 2^2 = 6$$

$$\tilde{q}_2 = a_2 - \frac{(a_2^T \tilde{q}_1)}{\|\tilde{q}_1\|^2} \tilde{q}_1 = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} - \frac{\begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}}{6} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} - \frac{0}{6} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad \|\tilde{q}_2\|^2 = 1^2 + (-1)^2 + 0^2 = 2$$

$$\tilde{q}_3 = a_3 - \frac{(a_3^T \tilde{q}_1)}{\|\tilde{q}_1\|^2} \tilde{q}_1 - \frac{(a_3^T \tilde{q}_2)}{\|\tilde{q}_2\|^2} \tilde{q}_2$$

$$= \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} - \frac{\begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}}{6} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - \frac{\begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} \cdot \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} - \frac{9}{6} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} - \begin{bmatrix} 3/2 \\ 3/2 \\ 3 \end{bmatrix} - \begin{bmatrix} 1/2 \\ -1/2 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \quad \|\tilde{q}_3\|^2 = (-1)^2 + (-1)^2 + 1^2 = 3$$

$$\Rightarrow q_1 = \frac{\tilde{q}_1}{\|\tilde{q}_1\|} = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} / \sqrt{6} = \begin{bmatrix} 1/\sqrt{6} \\ 1/\sqrt{6} \\ 2/\sqrt{6} \end{bmatrix}$$

$$q_2 = \frac{\tilde{q}_2}{\|\tilde{q}_2\|} = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} / \sqrt{2} = \begin{bmatrix} 1/\sqrt{2} \\ -1/\sqrt{2} \\ 0 \end{bmatrix}$$

$$q_3 = \frac{\tilde{q}_3}{\|\tilde{q}_3\|} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix} / \sqrt{3} = \begin{bmatrix} -1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \end{bmatrix}$$

Problem 5. (18 points) True or False (no explanation needed).

(a) If the columns of a matrix are independent, so are the rows.

False $A = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$

(b) If $A^T = -A$ then the row space of A equals the column space of A .

True

(c) The nullspace $N(A)$ and the row space $C(A^T)$ are orthogonal complements.

True

(d) If P is a projection matrix, then $P^2 = P$.

True

(e) The determinant of $AB - BA$ is zero.

False $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}, B = \begin{bmatrix} 0 & 0 \\ 0 & 1 \end{bmatrix}$
 $\det(AB - BA) = \det \begin{bmatrix} 0 & 2 \\ -3 & 0 \end{bmatrix} = 6 \neq 0$

(f) The area of the triangle formed by the three points $(-1, 2), (0, 5), (3, 3)$ is equal to $\frac{1}{2}\det A$

with $A = \begin{bmatrix} -1 & 2 & 1 \\ 0 & 5 & 1 \\ 3 & 3 & 1 \end{bmatrix}$.

False

Since $\det A = \begin{vmatrix} -1 & 2 & 1 \\ 0 & 5 & 1 \\ 3 & 3 & 1 \end{vmatrix}$
 $= (-1) \cdot (2) + 3 \cdot (-3) = -11$

not counter-clockwise ordered!